# 이석영 (Seok Young Lee), Ph.D.

세종대학교 지능정보융합학과
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## SUMMARY

Seok Young Lee received his B.S. degree in the Department of Electrical Engineering at Pohang University of Science and Technology (POSTECH), Pohang, South Korea, in 2011, and the Ph.D. degree in the Department of IT Convergence Engineering (ITCE) at POSTECH, Pohang, South Korea, in 2018. From 2018 to 2019, he was a Staff Engineer at Samsung Electronics. From 2019 to 2025, he was an Assistant Professor in the Department of Electronic Engineering at Soonchunhyang University. Since 2025, he has been with the Department of Artificial Intelligence and Information Technology at Sejong University, where he is currently an Associate Professor and leads the Intelligent Control and Robotics (ICaR) Laboratory. His research interests include robust control theory, delayed systems, networked control systems, robot manipulators, and embedded systems.

## INTEREST AREAS

**Control theory**: robust control, networked control, sampled-data control, neural-network-based systems, time-delay systems
**Robotics**: robot manipulators, adaptive sliding mode control, force/admittance control, multi-robot systems
**Embedded systems**: real-time implementation, motor control, sensor systems, on-device AI, signal processing

## EDUCATION

**Ph.D. IT Convergence Engineering (ITCE)** Mar. 2011 – Feb. 2018
*POSTECH, Pohang, Gyeongbuk, South Korea*

- Dissertation : Stability Analysis of Systems with Time-varying Delays via Slack Matrix based Approaches
- Advisor: Prof. Poogyeon Park

**B.S. Electrical Engineering (EE)** Mar. 2007 – Feb. 2011
*POSTECH, Pohang, Gyeongbuk, South Korea*

- Thesis : Teleoperated robot arm based on ATmega128

**High School Graduation** Mar. 2004 – Feb. 2007
*Gwangju Science Academy, Gwangju, South Korea*

## WORK EXPERIENCE

**Associate Professor** Sep. 2026 – Current
*Sejong University, Seoul, Republic of Korea*

*Assistant Professor* *Mar. 2025 – Aug. 2026*

- Department of Artificial Intelligence and Information Technology
  - Lectures: Probability and statistics, C programming, engineering mathematics, embedded systems, linear systems, nonlinear systems, etc.
- Intelligent Control and Robotics (ICaR) Laboratory
  - 3 Graduate students (1 Ph.D. candidate, 2 M.S. candidates)

**Assistant Professor** Sep. 2019 – Feb. 2025

*Soonchunhyang University, Asan, Chungnam, Republic of Korea*

- Department of Electronic Engineering
  - Lectures: Signal and system, automatic control, microprocessor, MATLAB programming, motor control, embedded programming, etc.

**Staff Engineer** Mar. 2018 – Aug. 2019

*Samsung Electronics, Hwaseong, Gyeonggi-do, Republic of Korea*

- Intelligent Control and Transfer Robot Laboratory, Mechatronics Team
  - Topics: Developments of servo drive, motion controller, S/W platform for control systems, vibration control algorithms, transfer robot, SCARA robot, rotary/direct-drive/linear motor control algorithms

## PUBLICATIONS

SCIE Journals **34** (Q1 **23**) · ESCI Journal **1** · International Conferences **5**

### SCIE Journal (* : Corresponding Author, based on JCR 2025)

### [2021 – Current]

[34] Hyojin Jo, Dogyun Park, **Seok Young Lee***, Nam Kyu Kwon*, "Feasibility and Evaluation of Effective Morphological Feature Representation for Contactless Blood Pressure Estimation Using Remote Photoplethysmography," Electronics, vol. 15, no. 17, p. 3864, Aug. 2026. (JCR Q2)
doi:10.3390/electronics15173864

[33] Sejik Oh, Bongjun Choi, **Seok Young Lee***, Nam Kyu Kwon*, "Adaptive Sliding Mode Control with Time-Delay Error Compensation and Admittance-Based Force Tracking," Mathematics, vol. 14, no. 13, p. 2323, Jul. 2026. (JCR Q1, top 5.6%)
doi:10.3390/math14132323

[32] **Seok Young Lee**, Nam Kyu Kwon*, "PD dynamic output-feedback admissibilization for descriptor systems with uniform input quantization," Communications in Nonlinear Science and Numerical Simulation, vol. 158, p. 109872, Jul. 2026. (JCR Q1, top 1.3%)
doi:10.1016/j.cnsns.2026.109872

[31] **Seok Young Lee**, Nam Kyu Kwon*, JunMin Park*, "Stability analysis and stabilization synthesis of asynchronously sampled-data systems via integral looped functionals composed of bivariate functions," Mathematics and Computers in Simulation, vol. 243, pp. 221-236, May 2026. (JCR Q1, top 1%)
doi:10.1016/j.matcom.2025.11.031

[30] **Seok Young Lee**, JunMin Park*, "Stabilization of Neutral Time-Delay Systems With Actuator Saturation via a Sampled-Data Control Approach," IEEE Access, vol. 13, pp. 178293-178301, 2025. (JCR Q2)
doi:10.1109/access.2025.3620850

[29] Jin Woong Lee, Jae Min Rho, Sun Gene Park, Hyuk Mo An, Minhyuk Kim*, **Seok Young Lee***, "Improved Adaptive Sliding Mode Control Using Quasi-Convex Functions and Neural Network-Assisted Time-Delay Estimation for Robotic Manipulators," Sensors, vol. 25, no. 14, p. 4252, Jul. 2025. (JCR Q2)
doi:10.3390/s25144252

[28] JunMin Park, **Seok Young Lee***, "Consecutive time-intervals-dependent looped functionals for stability analysis of linear systems with asynchronous sampling," International Journal of Robust and Nonlinear Control, vol. 35, no. 2, pp. 496-510, Jan. 2025. (JCR Q1, top 5.7%)
doi:10.1002/rnc.7658

[27] JunMin Park, **Seok Young Lee***, ”A novel looped functional for stability analysis of asynchronous sampled-data systems,” Asian Journal of Control, vol. 27, no. 1, pp. 41-50, Jan. 2025. (JCR Q3)
doi:10.1002/asjc.3547

[26] Hyuk Mo An, Jin Woong Lee, Dong Hee Seo, **Seok Young Lee***, ”An Adaptive Sliding Mode Control With Novel Sliding Variable-Based Adaptive Law for Disturbed Robot Manipulators,” IEEE Access, vol. 12, pp. 165227-165235, 2024. (JCR Q2)
doi:10.1109/access.2024.3493882

[25] Dong Hee Seo, Jin Woong Lee, Hyuk Mo An, **Seok Young Lee***, ”An Adaptive Sliding Mode Control Using a Novel Adaptive Law Based on Quasi-Convex Functions and Average Sliding Variables for Robot Manipulators,” Electronics, vol. 13, no. 19, Art. no. 3940, Oct. 2024. (JCR Q2)
doi:10.3390/electronics13193940

[24] **Seok Young Lee**, JunMin Park*, ”An enhanced looped-functional framework for stability analysis of sampled-data systems,” Journal of the Franklin Institute, vol. 361, no. 10, p. 106901, Jul. 2024. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2024.106901

[23] **Seok Young Lee**, JunMin Park*, ”Sampled-data stabilization for networked control systems under deception attack and the transmission delay,” Communications in Nonlinear Science and Numerical Simulation, vol. 131, p. 107817, Apr. 2024. (JCR Q1, top 1.3%)
doi:10.1016/j.cnsns.2024.107817

[22] JunMin Park, Nam Kyu Kwon, **Seok Young Lee***, ”Extended affine Bessel summation inequalities: Applications to stability analysis of linear discrete-time systems with time-varying delays,” Applied Mathematics and Computation, vol. 451, p. 128025, Aug. 2023. (JCR Q1, top 4.2%)
doi:10.1016/j.amc.2023.128025

[21] Minjae Park, **Seok Young Lee**, Jin Seok Hong, Nam Kyu Kwon*, ”Deep Deterministic Policy Gradient-Based Autonomous Driving for Mobile Robots in Sparse Reward Environments,” Sensors, vol. 22, no. 24, Art. no. 9574, Dec. 2022. (JCR Q2)
doi:10.3390/s22249574

[20] Nam Kyu Kwon, **Seok Young Lee***, ”An Affine Integral Inequality of an Arbitrary Degree for Stability Analysis of Linear Systems With Time-Varying Delays,” IEEE Access, vol. 9, pp. 51958-51969, 2021. (JCR Q2)
doi:10.1109/access.2021.3070149

## [2014 – 2020]

[19] Nam Kyu Kwon, **Seok Young Lee***, ”Novel Equalities for Stability Analysis of Asynchronous Sampled-Data Systems,” IEEE Access, vol. 8, pp. 177195-177205, 2020. (JCR Q2)
doi:10.1109/access.2020.3026736

[18] **Seok Young Lee**, Nam Kyu Kwon*, ”Proportional-Derivative State-Feedback Control for Singular Systems With Input Quantization,” IEEE Access, vol. 8, pp. 160065-160069, 2020. (JCR Q2)
doi:10.1109/access.2020.3020069

[17] JunMin Park, **Seok Young Lee**, PooGyeon Park*, ”A Less Conservative Stability Criterion for Discrete-Time Lur’e Systems With Sector and Slope Restrictions,” IEEE Transactions on Automatic Control, vol. 64, no. 10, pp. 4391-4395, Oct. 2019. (JCR Q1, top 12.1%)
doi:10.1109/tac.2019.2899079

[16] **Seok Young Lee**, JunMin Park, PooGyeon Park*, ”Bessel summation inequalities for stability analysis of discrete-time systems with time-varying delays,” International Journal of Robust and Nonlinear Control, vol. 29, no. 2, pp. 473-491, Jan. 2019. (JCR Q1, top 5.7%)
doi:10.1002/rnc.4398

[15] JunMin Park, **Seok Young Lee**, PooGyeon Park*, "An improved fragmentation approach to sampled-data synchronization of chaotic Lur'e systems," Nonlinear Analysis: Hybrid Systems, vol. 29, pp. 333-347, Aug. 2018. (JCR Q1, top 4.2%)
doi:10.1016/j.nahs.2018.02.006

[14] JunMin Park, **Seok Young Lee**, PooGyeon Park*, "An improved stability criteria for neutral-type Lur'e systems with time-varying delays," Journal of the Franklin Institute, vol. 355, no. 12, pp. 5291-5309, Aug. 2018. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2018.05.014

[13] Won Il Lee, **Seok Young Lee**, PooGyeon Park*, "Affine Bessel-Legendre inequality: Application to stability analysis for systems with time-varying delays," Automatica, vol. 93, pp. 535-539, Jul. 2018. (JCR Q1, top 14.8%)
doi:10.1016/j.automatica.2018.03.073

[12] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "Orthogonal-polynomials-based integral inequality and its applications to systems with additive time-varying delays," Journal of the Franklin Institute, vol. 355, no. 1, pp. 421-435, Jan. 2018. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2017.11.011

[11] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "Polynomials-based summation inequalities and their applications to discrete-time systems with time-varying delays," International Journal of Robust and Nonlinear Control, vol. 27, no. 17, pp. 3604-3619, Nov. 2017. (JCR Q1, top 5.7%)
doi:10.1002/rnc.3755

[10] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "Stability analysis of discrete-time systems with time-varying delays: Generalized zero equalities approach," International Journal of Robust and Nonlinear Control, vol. 27, no. 6, pp. 981-999, Apr. 2017. (JCR Q1, top 5.7%)
doi:10.1002/rnc.3613

[09] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "Polynomials-based integral inequality for stability analysis of linear systems with time-varying delays," Journal of the Franklin Institute, vol. 354, no. 4, pp. 2053-2067, Mar. 2017. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2016.12.025

[08] Won Il Lee, **Seok Young Lee**, PooGyeon Park*, "A combined reciprocal convexity approach for stability analysis of static neural networks with interval time-varying delays," Neurocomputing, vol. 221, pp. 168-177, Jan. 2017. (JCR Q1, top 20.2%)
doi:10.1016/j.neucom.2016.09.074

[07] **Seok Young Lee**, Won Il Lee, PooGyeon Park, "Improved stability criteria for linear systems with interval time-varying delays: Generalized zero equalities approach," Applied Mathematics and Computation, vol. 292, pp. 336-348, Jan. 2017. (JCR Q1, top 4.2%)
doi:10.1016/j.amc.2016.07.015

[06] PooGyeon Park*, **Seok Young Lee**, Won Il Lee, "New stability analysis for discrete time-delay systems via auxiliary-function-based summation inequalities," Journal of the Franklin Institute, vol. 353, no. 18, pp. 5068-5080, Dec. 2016. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2016.07.011

[05] Won Il Lee, **Seok Young Lee**, PooGyeon Park*, "A combined first- and second-order reciprocal convexity approach for stability analysis of systems with interval time-varying delays," Journal of the Franklin Institute, vol. 353, no. 9, pp. 2104-2116, Jun. 2016. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2016.03.017

[04] PooGyeon Park*, Won Il Lee, **Seok Young Lee**, "Auxiliary function-based integral/summation inequalities: Application to continuous/discrete time-delay systems," International Journal of Control, Automation and Systems, vol. 14, no. 1, pp. 3-11, Feb. 2016. (JCR Q2)
doi:10.1007/s12555-015-2002-y

[03] Won Il Lee, **Seok Young Lee**, PooGyeon Park*, "Improved stability criteria for recurrent neural networks with interval time-varying delays via new Lyapunov functionals," Neurocomputing, vol. 155, pp. 128-134, May 2015. (JCR Q1, top 20.2%)
doi:10.1016/j.neucom.2014.12.040

[02] PooGyeon Park*, Won Il Lee, **Seok Young Lee**, "Auxiliary function-based integral inequalities for quadratic functions and their applications to time-delay systems," Journal of the Franklin Institute, vol. 352, no. 4, pp. 1378-1396, Apr. 2015. (JCR Q1, top 13.5%)
doi:10.1016/j.jfranklin.2015.01.004

[01] Won Il Lee, **Seok Young Lee**, PooGyeon Park*, "Improved criteria on robust stability and $H_\infty$ performance for linear systems with interval time-varying delays via new triple integral functionals," Applied Mathematics and Computation, vol. 243, pp. 570-577, Sep. 2014. (JCR Q1, top 4.2%)
doi:10.1016/j.amc.2014.05.116

### ESCI/Scopus Journal (* : Corresponding Author, based on JCR 2025)

[01] Sejik Oh, Kyoung Min Lee, **Seok Young Lee**, Nam Kyu Kwon*, "Movement Direction Classification Using Low-Resolution ToF Sensor and LSTM-Based Neural Network," Journal of Sensor and Actuator Networks, vol. 14, no. 3, p. 61, Jun. 2025. (JCR Q2)
doi:10.3390/jsan14030061

### International Conference (* : Corresponding Author)

[05] **Seok Young Lee***, "Improved stability criteria for sampled data systems via a novel looped-functional," 2022 13th Asian Control Conference (ASCC), 2022.

[04] **Seok Young Lee**, Min Su Kim, Chaneun Park, PooGyeon Park*, "Analysis on stability for generalized neural networks with time-varying delays via second-order orthogonal polynomials-based integral inequality," 17th International Conference on Control, Automation and Systems (ICCAS), Jeju, Korea, pp. 432-436, Oct. 2017.
doi:10.23919/iccas.2017.8204478

[03] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "Improved slack-matrix-based summation inequality and applications to discrete-time systems with time-varying delays," IEEE 55th Conference on Decision and Control (CDC), Las Vegas, NV, USA, pp. 2196-2200, Dec. 2016.
doi:10.1109/cdc.2016.7798589

[02] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "Combined-slack-matrix-based integral inequality: Application to time-delay systems," 55th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE), Tsukuba, Japan, pp. 1564-1568, Sep. 2016.
doi:10.1109/sice.2016.7749198

[01] **Seok Young Lee**, Won Il Lee, PooGyeon Park*, "New stability criteria for linear systems with interval time-varying delays via an extended state vector," 10th Asian Control Conference (ASCC), Kota Kinabalu, Malaysia, pp. 1-6, May 2015.
doi:10.1109/ascc.2015.7244650

## RESEARCH ACTIVITIES

**Resilient/Robust Control for Modular Manipulator Reassembly Systems** Mar. 2026 – Feb. 2031
*NRF Korea, Young Researcher Program (Type B) – Principal Investigator*

- An integrated resilient and robust control framework for reassembly systems using modular heterogeneous manipulators

**Training Program for AX-based Manufacturing Process Technology** Mar. 2026 – Feb. 2031
*Korea Institute for Advancement of Technology (KIAT) – Co-Investigator*

- On-device AI manufacturing platform

**Research Curriculum on Next-generation Mobility Communications** Jul. 2024 – Feb. 2025
*IITP – Co-Investigator*

**Networked Control Theory with Application to Platoon Mobility in V2X Environments** Jun. 2024 – Jan. 2025
*Daejeon–Sejong–Chungnam Regional Innovation Platform – Principal Investigator*

**RNA Innovative Talent Development Program** Jan. 2023 – Feb. 2024
*IITP – Co-Investigator*

**Robust Sampled Data Control for Networked Systems with Uncertainty** Mar. 2020 – Feb. 2023
*NRF Korea, First Research Grant – Principal Investigator*

- Stability analysis and controller synthesis of networked systems
- Developments of sampled-data control algorithms

**MPC Design for Time-delay Systems using Relaxation Techniques** Sep. 2019 – Aug. 2020
*Soonchunhyang University*

- Stability analysis and controller synthesis of time-delay systems
- Developments of model predictive controller (MPC)

**Motor and Motion Controller for Industrial Robots** Mar. 2018 – Aug. 2019
*Samsung Electronics*

- Developments of S/W platform in servo drive and motion controller
- Developments of vibration control algorithms and robust control algorithms
- Robust control of transfer robot and SCARA robot
- Robust control of rotary motor, direct-drive motor and linear motor

**Robust Control of Electro-Hydraulic Excavator Automation System** Aug. 2014 – Oct. 2015
*Doosan Infracore*

- Modeling of electro-hydraulic system and system identification
- Developments of robust control algorithms including time delay controller and gain scheduling

**Localization Algorithm in Underwater Environment** Feb. 2012 – Nov. 2014
*KIOST*

- Developments of localization algorithms for underwater robot
- Direction and location estimation algorithms for sound sources by utilizing generalized cross correlation(GCC), particle filter, and recursive total least square (RTLS) algorithms

**Smart Camera for High-speed TFT-LCD Panel Inspection** Mar. 2011 – Feb. 2013
*Nbiz*

- Developments of smart camera for TFT-LCD panel inspection
- PCB artwork, developments of image processing algorithms and S/W platforms in embedded systems

## ACADEMIC ACTIVITIES

KIEE Member(2019-), ICROS Member(2017-)

## HONORS AND AWARDS

**Young Researcher Award (신진연구자상)** Oct. 2022
*The Korean Institute of Electrical Engineers (대한전기학회), Information and Control Society*

**Outstanding Paper** Feb. 2018
*Department of IT Convergence Engineering, Pohang University of Science and Technology*

**Undergraduate Fellowship (Merit-based)** Mar. 2007
*Department of Electrical Engineering, Pohang University of Science and Technology*

- Admission with the highest distinction
- Full tuition (granted for two semesters)